

CAPSTONE

LOGISTICS LANE COST OPTIMIZATION

Eskwelabs Innovation Fellowship Intern

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✿ ABOUT ME

WHO AM I?

- Fourth-year BS Mathematics student at UPD
- Classical music and violin enthusiast
- Past internship experiences in Data Science

WHAT AM I INTERESTED IN?

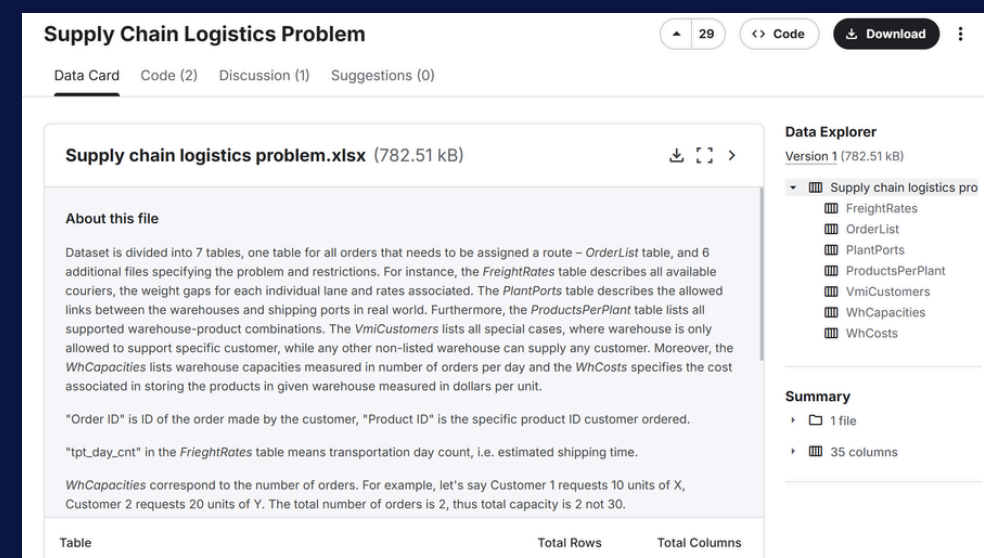
- Highly interested in both theoretical and applied math, especially in its application in the business setting.
- Interested in answering the whys and hows of things and also doing deep dives to find the answer.



MY EIF JOURNEY

LOGISTICS LANE COST OPTIMIZATION

- 1 Based on the Supply Chain Logistics Problem dataset
- 2 User has a lot of freedom to input parameters and distributions.



PROJECT PORTFOLIO

PROBLEM/CONTEXT

Supply chain or operations analysts want to test optimization models at different network scales, demand volatilities, or constraint densities, however:

- 1 There may be instances where edge cases come up very rarely in their current datasets.
- 2 The optimizer must find the allocation that minimizes total cost subject to hard network constraints.

Thus, a synthetic dataset generator that gives analysts extensive freedom has been created.



PROJECT PORTFOLIO

ROLE AND CONTRIBUTIONS

ORIGINAL DATASET

- Searched the internet for an existing dataset that would tackle the assigned problem.

ORIGINAL DATASET EDA

- Performed EDA using Python (Pandas) and Power BI to identify relationships, trends, distributions, and anomalies.

PROMPT ENGINEERING FOR SYNTHETIC DATA GENERATION

- Created and optimized prompts to achieve an efficient algorithm that creates the desired synthetic dataset.



PROJECT PORTFOLIO

ROLE AND CONTRIBUTIONS

SYNTHETIC DATASET EDA

- Performed EDA on the synthetic dataset and performed statistical methods to validate user-input distributions.

IMRAD PAPER

- Created the IMRAD Paper to summarize methodology and results.

DATA DICTIONARY

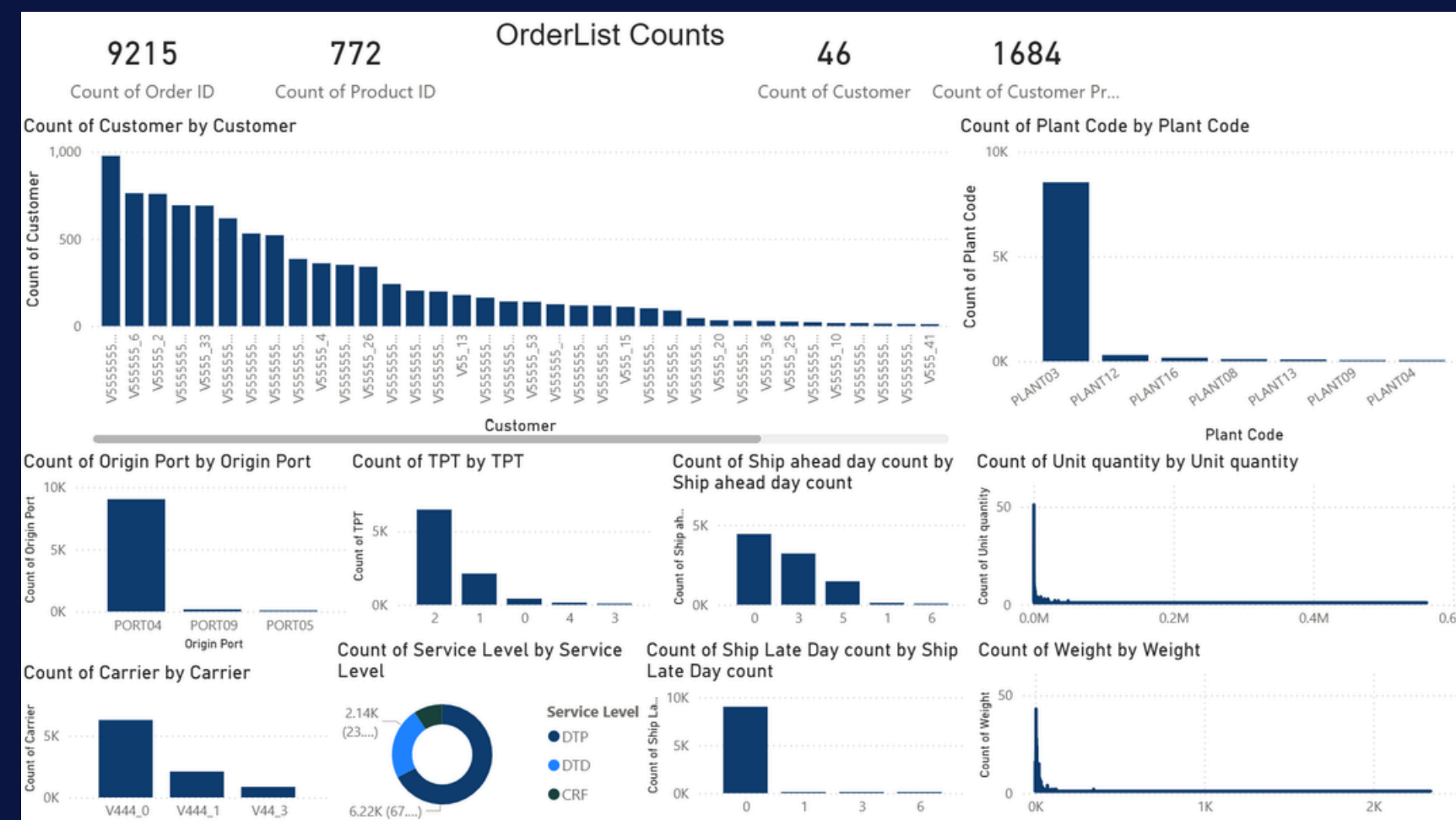
- Created the data dictionary to have an easy reference to the feature definitions.



PROJECT PORTFOLIO

OUTPUT/DELIVERABLES

DASHBOARD OF ORDERS LIST IN ORIGINAL DATASET



PROJECT PORTFOLIO

OUTPUT/DELIVERABLES

INPUT FREEDOM

19 VALIDATION CHECKS

**Synthetic
Dataset
Generator**

CLICK HERE!

```
@dataclass
class GeneratorConfig:
    """Full configuration for the supply chain synthetic data generator."""

    # --- Scale ---
    n_orders: int = 10_000
    n_customers: int = 500
    n_products: int = 200
    n_plants: int = 20
    n_ports: int = 8
    n_carriers: int = 12

    # --- Time range ---
    start_date: str = "2026-01-01"
    end_date: str = "2026-01-02"

    # --- VMI settings ---
    # VMI plant IDs must be valid PLANT_XXX labels within n_plants.
    # Pass None to auto-select ~20 % of plants as VMI.
    vmi_plant_ids: Optional[Tuple[str, ...]] = None
    vmi_customer_ratio: float = 0.15 # fraction of customers that are VMI

    # --- Product-plant coverage ---
    min_products_per_plant: float = 0.30 # each plant carries ≥30 % of catalogue
    max_products_per_plant: float = 0.70 # each plant carries ≤70 % of catalogue

    # --- Capacity buffer ---
    # Plant daily capacity = peak assigned quantity × (1 + buffer)
    capacity_buffer: float = 0.20

    # --- Order quantity distribution ---
    # "uniform" | "lognormal" | "zipf"
    order_qty_dist: str = "uniform"
    min_order_qty: int = 5
    max_order_qty: int = 50
```

Running validation ...

✓ All 19 validation checks passed.

OrderList	:	10,000 rows
FreightRates	:	1,610 rows
WhCosts	:	20 rows
WhCapacities	:	20 rows
ProductsPerPlant	:	1,932 rows
VmiCustomers	:	113 rows
PlantPorts	:	42 rows

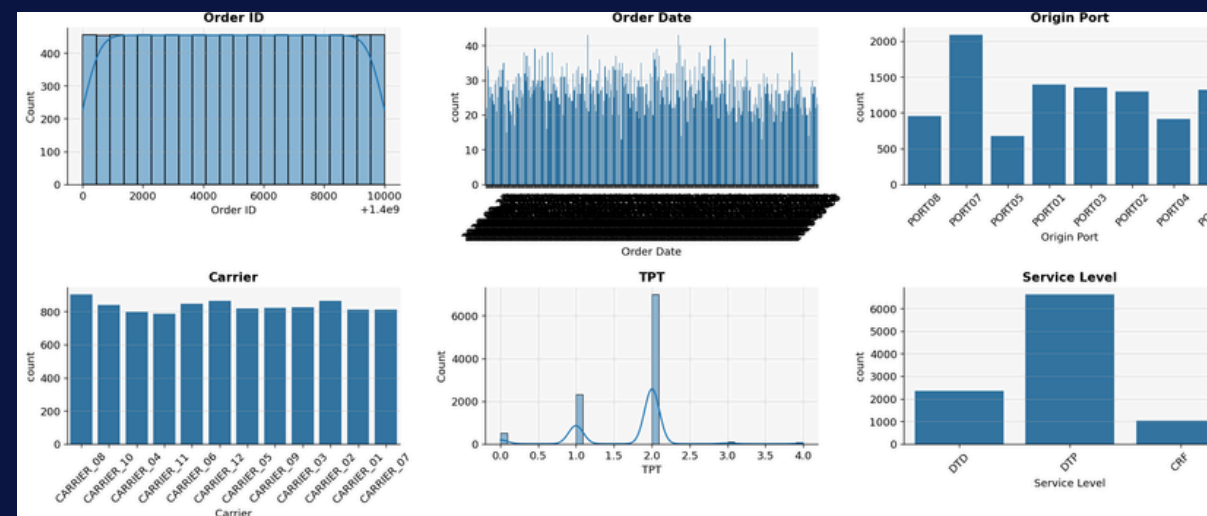
PROJECT PORTFOLIO

OUTPUT/DELIVERABLES

**Synthetic
Dataset
EDA**

CLICK HERE!

DISTRIBUTION CHECKS



STATSTICAL TESTS

Kruskal-Wallis H-test: freight rate differences across transport modes
H = 1233.277, p-value = 1.58e-268
→ Rates differ significantly across modes

Mean rates (\$/kg):

AIR : \$2.171
SEA : \$0.131
GROUND: \$0.331

Real-world benchmarks (approximate):

AIR: \$1.50-\$5.00/kg
SEA: \$0.05-\$0.30/kg
GROUND: \$0.10-\$0.60/kg

Spearman correlation (rate vs transit days): $r = -0.823$, $p = 0.0000$
Expected: negative – faster = more expensive

PROJECT PORTFOLIO

OUTPUT/DELIVERABLES

IMRAD
Paper

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ZERO ASSUMPTION VIOLATIONS

RESULTS

3.1 Referential Integrity and Constraint Compliance

The validation suite confirmed zero violations across all five constraint checks: plant-port combinations, plant-product combinations, daily warehouse capacity, carrier coverage, and VMI routing rules. This result is expected by construction: the generator enforces these constraints during row assembly rather than checking them post-hoc. The daily peak load per plant never exceeded the assigned capacity; the maximum observed peak was 8 orders per day against capacities ranging from 10 to 99 orders per day.

CAN MIMIC REAL-LIFE SCENARIOS

3.7 Cost Estimation

For orders successfully matched to a freight rate row (97.3% of orders), the freight cost per order was computed as:

```
freight_cost_i = max(min_cost, rate * Weight_i * qty_i) (30)
```

Total cost per order was the sum of freight and warehouse costs. The mean freight cost was \$492.10 per order; the mean warehouse cost was \$24.80 per order, meaning freight accounted for approximately 95% of the total cost. This ratio is consistent with international supply chains where transport dominates storage cost.

✿ PROJECT PORTFOLIO

SKILLS DEMONSTRATED

- Python
- Power BI
- Pandas
- Object-Oriented Programming
- Exploratory Data Analysis
- Statistical Methods
- Data Science



CAPSTONE REFLECTION

WHAT I'VE LEARNED

I have learned a great deal not only from the project my teammate and I completed, but also from the events I have attended. I have learned that creating synthetic datasets is difficult, especially when I want them to offer user freedom and still mimic real-life scenarios. During the actual coding part of the algorithm, my teammate and I encountered some hurdles where we were stuck and unsure how to even begin coding this. However, from the events that I was able to attend, I have learned that with the current capabilities of AI, almost anything can be built as long as the prompts are optimized. Thus, I leveraged its power in order to create the algorithm.



CAPSTONE REFLECTION

WHAT I'M PROUD OF

I am extremely proud that I was able to even finish this project, even if I was just able to tackle one. The dataset I have chosen actually made the synthetic data generation part very difficult on our end, given that the dataset has lots of tables, lots of hidden assumptions, and some anomalies. I am also proud that we were able to create an algorithm that works pretty well. Not only does it have user freedom, where the user can have a lot of control over the parameters and distributions, but it also has tons of validity checks to ensure that the resulting dataset mimics real-life scenarios.



CAPSTONE REFLECTION

WHAT I WOULD IMPROVE

Although the generator looks good, there are still specific combinations of values that would result in infeasibility or invalid results. What I'd improve is maybe I would make sure that a dataset is guaranteed to be produced, but a score that depends on feasibility and validity would be added. I would also improve my exploratory data analysis skills since I have encountered problems in that part as well, where we thought we were already good to go in terms of creating the generator, but it turns out that there was an assumption we failed to consider because we missed that part during the exploratory data analysis. I would also improve the weight brackets. Currently the brackets are all the same for each carrier, but that is certainly not the case in the original dataset.



LOOKING FORWARD

ROLES/OPPORTUNITIES I'M EXPLORING NEXT

This experience has been, without a doubt, extremely invaluable. I was able to learn a subfield of data science that I was not really putting much focus on before I applied to this internship. My next step would actually be to start applying the extensive skills that I've developed from my past internship to actual work. I have started applying to data science and data analytics roles, and I have already gotten quite several interviews. I am really hoping that my jobhunting journey ends well.



THANK YOU!



MARIANO, Isaiah John L

